A Convex Analysis of a Soft Capture and Avoidance Problem for Spacecraft Docking

Soft Capture and Avoidance Problem Convexified

Aidan R. Giebelhausen

Purdue University

Course Number: AAE 590 ACA

Course Instructor: Professor Kenshiro Oguri

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1. Problem Statement

1.1 Project Introduction

This project is an important part of a spacecraft maneuver set and it is something that most spacecraft will need to be doing in the future. As humans advance as a civilization, new ideas and projects for space stations and robust spacecraft missions are being developed. As these missions grow in complexity, the need for reliable docking methods becomes increasingly relevant. Thus, this project is extremely important to consider, as it is important to maintain safety and a measure of control when docking. This also brings about the point of relevant challenges, as a soft capture problem has many factors to consider.

This report will first delve into a thrust and velocity bound to ensure spacecraft safety as if it is arriving too fast it could be dangerous. It will then discuss a docking port angle constraint to make sure that the chaser is within a cone of a certain half angle. The chaser spacecraft also must not collide with the target spacecraft during flight. This problem will also be trying to minimize fuel consumption under a bang bang profile. This setup will lead to many nonconvexities which will need to be analyzed, discussed and fixed with change of variables and reforming constraints, which will be another challenge. Then there is the implementation and analysis of the problem and the visual representation of it working. This will be challenging as well because the entry angles and docking procedure will have to be visualized someway.

1.2 Similar Projects

The main inspiration for this project was a paper by I. S. Sow, G. Gutow, H. Choset and Z. Manchester. They published a paper on a similar problem which was titled “A Convex Formulation of the Soft-Capture Problem”. That paper was extremely enlightening in creating and developing this problem.

It will be referred to as Sow's paper for readability. Sow's paper was an analysis of the same problem as this paper but with different objectives and a larger scope. This paper will delve more into the convex constraints themselves and how they are formulated one by one and applied then all together. It will be about how each of those constraints change the problem. Sow's paper was a very in-depth look at all of those constraints combined, and the feasibility of it. This paper has the benefit of applying the fact that all of those constraints are confirmed to work together. Due to time constraints, some of these constraints will not be applied as they are in Sow's paper due to the shear complexity. Otherwise, some other papers will be helpful and were instrumental. These papers are "Fast trajectory optimization via successive convexification for spacecraft rendezvous with integer constraints", and "A convex optimization approach for finite-thrust time-constrained cooperative rendezvous". These are all in the references section below. Both of these papers helped in the convexification of each separate constraint because of differences when you do them all together versus one at a time.

2. Objectives

The project will be in three dimensions exclusively as that is much more applicable than two dimensions. Each constraint will be visualized and analyzed to make certain that they are working properly and can be applied separately. All constraints can be used separately in different real-world problems. It will break down like that so it can be analyzed to check if all parts of the problem are working properly. So, milestone one will be set up and convexity of the entire problem separately done. This will also work towards the goal of understanding the problem as each constraint can still be applied separately in terms of avoidance, minimum fuel, or angle lineup of spacecraft. This means it will be a good first level that will be able to be checked but will still advance the project. To further discuss and check on progress it will be separated into smaller categories like those ones above, where they can all be plotted and done as separate problems. This will visualize for the reader and increase understanding of the problem and if they are working it will be a good milestone two. Then they will all be recombined on paper and convexified together into one problem. Once that is completed that will be milestone three. Then they will all be put into CVX and the entire problem will be visualized at different angles and constraints and metrics. This will further increase understanding of the problem and what is feasible. It will also be a good milestone four. Then this will lead to an analysis of all of these results and formulation into a final document as milestone five. Now for a stretch result, as a milestone six, it will be an analysis of this problem with multiple spacecraft around the capture spacecraft some of which could be moving faster relative to the capture spacecraft. This will lead to more constraints and complexity. These categories will be made into a list and checked as metrics of success.

The levels will be shown below in Table 1 with information.

Level	Description and Goals
One	• Problem formulation - Explain all of the constraints and problems that will be addressed. • Problem formulation - Establish values for the problem and define it completely. • Convexification of every part of the problem as separately as possible at first. • Only the one capture spacecraft in the relative space. • Check one equation at a time to make sure convexity is held and try to implement each constraint on the problem separately in CVX. • Use linearized orbital dynamics. (CWH) • Visualization of all separate parts of the problem to check if feasible and working.
Two	• Combining and refining the entire convex problem. • Implementation of all of the constraints in convex programming. • Visualization of all parts of the problem. • Analysis of results gathered.
Three	• Analysis of docking procedure when multiple other spacecraft are around the capture spacecraft. • Simple relatively moving spacecraft as one of the ones around the area as well, so a dynamic constraint. • Convexification of those parameters. • CVX software analysis of the newer complex problem.

Table 1: Levels and Goals

3. Problem Definition and Methodology

3.1 Assumptions and Relative Dynamics

There are five key assumptions that will be made about this problem which will be shown in Table 2.

Number	Assumption
1	Chief orbit is circular and chaser operates in a local-vertical local-horizontal frame.
2	The relative motion follows the linearized Clohessy-Wiltshire-Hill (CWH) dynamics.
3	Only one chaser spacecraft is involved in the docking maneuver.
4	The final docking location is fixed and aligned with the target docking port.
5	All constraints are deterministic and can be modeled separately.

Table 2: Key Assumptions

As stated previously, this problem will be solved by applying each constraint separately, so in this section, each constraint of level one will be discussed on its own. This means it will be separated by headers for each different part and the methodology and any extra discussion on how each thing was done will be in its relative part. Each section will still share the same boundary conditions and objective function.

This problem will be done following CWH (Clohessy–Wiltshire equations) dynamics. The CWH equations are shown in equation 1. The chaser's position will be given by $x(t) = [r_1(t), r_2(t), r_3(t)]'$ and velocity $v(t) = [\dot{r}_1(t), \dot{r}_2(t), \dot{r}_3(t)]'$. The variable n is the mean motion of the orbit and u is defined by $u(t) = [u_x(t), u_y(t), u_z(t)]'$. The state space form is shown in equation 2 as well as the A and B matrix. These equations will be used to increase the feasibility of dealing with the problem in the amount of time given and the ability to convexify the problem. They are a linear problem and make key assumptions in

the orbit that will make it linear and simpler than a nonlinear form. These assumptions are circular orbit and the orbits are close enough that the difference between them is negligible in calculations.

$$\begin{aligned}\ddot{x} &= 3n^2x + 2n\dot{y}\\ \ddot{y} &= -2n\dot{x}\\ \ddot{z} &= -n^2z\end{aligned}$$

Equation 1: Clohessy–Wiltshire equations

$$\dot{x} = Ax + Bk$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 & 0 & 0\\ 0 & 0 & 0 & 0 & 1 & 0\\ 0 & 0 & 0 & 0 & 0 & 1\\ 3n^2 & 0 & 0 & 0 & 2n & 0\\ 0 & 0 & 0 & -2n & 0 & 0\\ 0 & 0 & -n^2 & 0 & 0 & 0 \end{bmatrix} \boldsymbol{B} = \begin{bmatrix} 0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\\ 1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1 \end{bmatrix}$$

Equation 2: State Space Form and A and B Matrix

3.2 Objective, Minimum Fuel Transfer, and Dynamics Convexification

This problem at a fundamental level will be a minimum fuel transfer. The goal is to minimize the integral of the norm of u(t) over time while maintaining safety parameters. This is going to be discretized because of the fact that the integral is not a convex function. When it is in the discretized form however, it is and convex solvers inherently cannot solve finite dimensional problems, they need the variables to be defined at a finite number of points. This is why it is discretized into a time grid k. This form is shown

in Equation 3 and is convex. This form will be applied to each separate constraint, and if it needs to be changed it will be mentioned in that section. There is also a need to discretize equation 2 to make sure everything is linear and convex; this is shown in equation 4.

$$min \sum_{k=1}^{N} \|u_k\|_2$$

Equation 3: Discretized Objective Function

$$\dot{x}_{k+1} = A_k x_k + B_k u_k$$

Equation 4: System Dynamics Convexified

3.3 Thrust Magnitude Bound Constraint

To ensure safe and physically realizable spacecraft maneuvers, the control input must respect a hard limit on the available thrust. This constraint models the propulsion system's maximum output and prevents unsafe accelerations during the docking approach. The maximum thrust constraint will be described as $||u_k||_2 <= U_{max}$. This is the discretized form at each point k same as above for 3.2 and what the velocity and every other constraint will be in as well. This constraint is deterministic and will likely be a known safety or hardware value. For this problem it will be set to 0.0001 m/s^2. This is to see the changes with an extremely low input from the thrusters, this will definitely have to be changed as the problem becomes more complex it was already workshopped quite a bit to find this value and the boundary conditions that it worked with even for just this constraint. This is a convex constraint so the only problem is making sure it is feasible.

3.4 Maximum Velocity Constraint

An important safety consideration in spacecraft docking is limiting the relative velocity of the chaser with respect to the target. High approach speed increases the risk of collision or damage upon contact, particularly in soft-capture scenarios where the structure could be damaged by a fast chaser

coming into contact. To ensure a smooth and controlled approach, the magnitude of the relative velocity constraint will be described as $||v_k||_2 <= V_{max}$. This V_{max} value will be similarly described like U_{max} where it is deterministic and a safety or hardware value. For this problem the value will be set to 0.031 m/s. This works because the norm of velocity starting is 0.03 m/s so a velocity constraint right above this works best for interesting results but as the problem becomes more complex it might have to change. This is a convex constraint so the only problem is making sure it is feasible.

3.5 Collision Constraint

The spacecraft needs a range of safety to protect itself and other potential spacecrafts from damage. This leads into the design of a predefined exclusion zone around the target of the maneuver. This constraint will be modeled as a keep in area where it cannot leave a certain zone where other spacecraft or debris could be. It will be represented as a sphere around the target spacecraft. This constraint will enforce a minimum separation distance at all times. This constraint can be seen as $||r_k||<= r_{safe}$. Where r_{safe} is a range where the spacecraft cannot go out of. For this project it will be a range of 30 meters and it will be worked with to see if it is feasible. This constraint will also be played with for a level three goal where there are specific another spacecraft to avoid. This will be modeled as a nonconvex problem since it would be $||r_k||>= r_{safe}$. To maintain convexity while enforcing collision avoidance, we approximate the spherical keep-out zone using a convex polyhedral region defined by a set of separating planes. This means that it would be a linear equation of something like $n'r_k \geq r_{safe}$ where n is a unit vector in order to make it linear. This will be discussed more in results and analysis depending on what can be accomplished. Upon working with the problem more, the spacecraft not being able to go out of a range is important but did not apply as much when this constraint was on its own so the linear equations will be the focus for the solo results. For this problem example I will use $r_{safe} = 5$, $r_{obs} =$ [15; 10; 5], so it is in the path of the object normally. The final value is $\hat{n}$, which is shown in equation 5, this is using a previous optimal path. This means that in the code, a convex problem will be run without

the constraint, then it will run with it and at each point, the previous route will be compared to the object and then compared to the current trajectory. This will lead to an accurate path constraint that goes on a more optimal path. These values mean that it will stay at least 5 meters away, for r_{safe}, the obstacle is at 15,10,5 for r_{obs}. The final convex input can be seen in Equation 6. This way of formatting is much simpler than the one done in most reference papers that have been read to do this project.

$$\hat{n} = \frac{r_{nom} - r_{obs}}{\|r_{nom} - r_{obs}\|}$$

Equation 5: $\hat{n}$ for Collision Constraint

$$\hat{n}^T * (x_k - r_{obs}) \geq r_{safe}$$

Equation 6: Collision Constraint

3.6 Field of View Constraint

This one will be the hardest constraint to apply by far. As discussed in the introduction, it is not sufficient to only constrain the final position of the spacecraft. It is often necessary for the spacecraft to arrive with proper alignment for docking mechanisms. It can also be used in specific station keeping of satellite missions where the alignment of the craft is of importance. This setup will follow the general setup that can be found in Sow[1] in the Field of View section. This was extremely helpful in setting up this problem as it was very difficult. First, there is going to be a new slack variable in the form of ρ_k which will also be added to the minimize line in CVX where the sum will be minimized. This slack variable is a lifted form of x and becomes a constraint in the form of Equation 7. There is an arduous process to get

[1] I. S. Sow, G. Gutow, H. Choset and Z. Manchester, *A Convex Formulation of the Soft-Capture Problem*, 2024 International Conference on Space Robotics (iSpaRo), Luxembourg, Luxembourg, 2024, pp. 131-138, doi: 10.1109/iSpaRo60631.2024.10687907.

to the next equation for the other constraint that is done in Sow. The first step is to set up the equation of attitude control with the new slack variables which becomes equation 8. This is pulled from Sow and is very interesting because it is done with some matrix tricks to get rid of the norms in the equation. Things like Equation 7 are turned into $r_k^T r_k \leq \rho^2$. This leads to the extended Equation 8. Phi is the variable for $\omega_{max}\Delta t > 0$. This is the angular tolerance between the line-of-sight vectors. Then, two new positive definite variables z_k and H are added which will simplify the equation and the final Equation 9 will be arrived at. Equations 7 and 9 are the ones that will be implemented in the code. Initial conditions are the same as normal with wmax = 0.05 and rmax = 30.

$$0 \leq \rho_k \leq r_{max}, \|r_k\|_2 \leq \rho_k$$

Equation 7: Constraints with Slack Variable

$$r_k^T r_k + r_k^T r_{k+1} + r_k^T r_{k+1} \leq \rho_k^2 - \rho_k \rho_{k+1} \cos(\phi) + \rho_{k+1}^2$$

Equation 8: Expansion of Quadratic Form from Sow

$$\left\|\sqrt{H} z_k\right\| \leq \sin\frac{\theta}{2} * (\rho_k + \rho_{k+1})$$

$$z_k = \begin{bmatrix} r_k \\ r_{k+1} \end{bmatrix}, \quad H = \begin{bmatrix} I_{3x3} & 0.5I_{3x3} \\ 0.5I_{3x3} & I_{3x3} \end{bmatrix} > 0$$

Equations 9: Final Field of View Constraint with Variables from Sow

3.7 Boundary Conditions and Time

The boundary conditions are an extremely important part because they can easily make the problem infeasible. In this problem it can be considered that the initial position and velocity and final position and velocity are known. This is because based on problem definition, velocity will be zero at the

end, and the point of arrival is known as the chief spacecraft docking port which will be considered as the origin. The initial position and velocity is something that will be set arbitrarily based on problem definition and feasibility. So, for now, $x_0 = [r_0; v_0]$ and $x_f = [r_f; v_f]$. $r_0 = [20; 20; 10]$, $v_0 = [-0.02;-0.02;-0.01]$, $r_f = [0;0;0]$, $v_f = [0;0;0]$. These values are all given in meters and meters per second. This is because this is a soft capture problem so the most important part is to do with close range docking maneuvers. This will still show interesting and helpful results as there can be extremely small u_{max} and v_{max} and it will be important to see how these affect the problem. Another important note is the final time and discretization node number. Final time is something that will change the entire problem since this problem will have a set value of discretization node number of 50. This means that every time final time is changed it will change the time step, which in turn will change the Ak and Bk matrixes, thus changing the entire problem. This is something that will be noted as it is needed but as a rule It will start at 1000 seconds since it should not take extremely long but there is time for maneuvers as this will be more informative. The A_k and B_k for this time step are shown in equation 10.

```
Ak =                                                                    Bk =

    1.0003         0         0   11.9996    0.1647         0   71.9989    0.6590         0
   -0.0000    1.0000         0   -0.1647   11.9985         0   -0.6590   71.9955         0
         0         0    0.9999         0         0   11.9996         0         0   71.9989
    0.0000         0         0    0.9999    0.0275         0   11.9996    0.1647         0
   -0.0000         0         0   -0.0275    0.9996         0   -0.1647   11.9985         0
         0         0   -0.0000         0         0    0.9999         0         0   11.9996
```

Equation 10: Ak and Bk at Time Step of One Hundred

3.8 All Constraints Combined as Much as Possible

It was planned to combine all of the constraints into one but that is an extremely difficult task which is outside of the scope of my abilities in this amount of time. Especially after doing the angle constraint. An in-depth analysis of this can be found in Sow[2]. This paper was extremely useful in

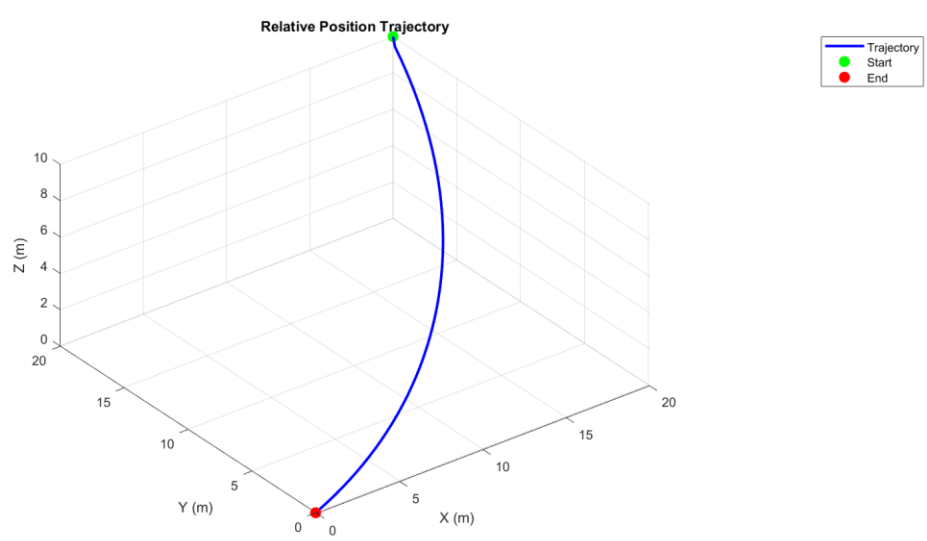
[2] I. S. Sow, G. Gutow, H. Choset and Z. Manchester, *A Convex Formulation of the Soft-Capture Problem*

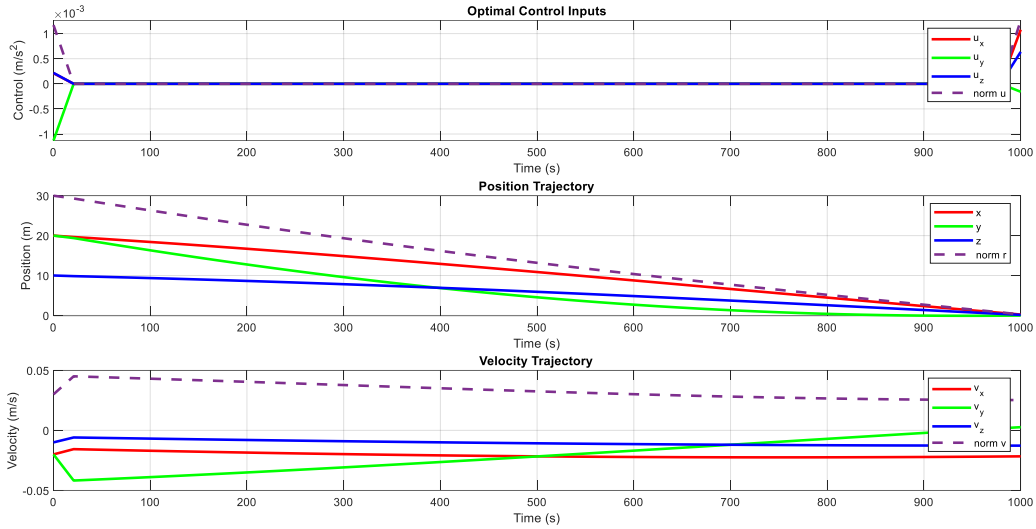
checking the work done in this paper especially on the application of the docking angle constraint. The problem of this constraint was attempted but was stopped on the convexification, without the report to follow, it was very difficult and it was thought it would not be quite as useful to just do what was done in Sow. This is why the paper will be referred to in the results and analysis. This paper is also starting to become very long with what analysis I wanted to have in here.

4. Results

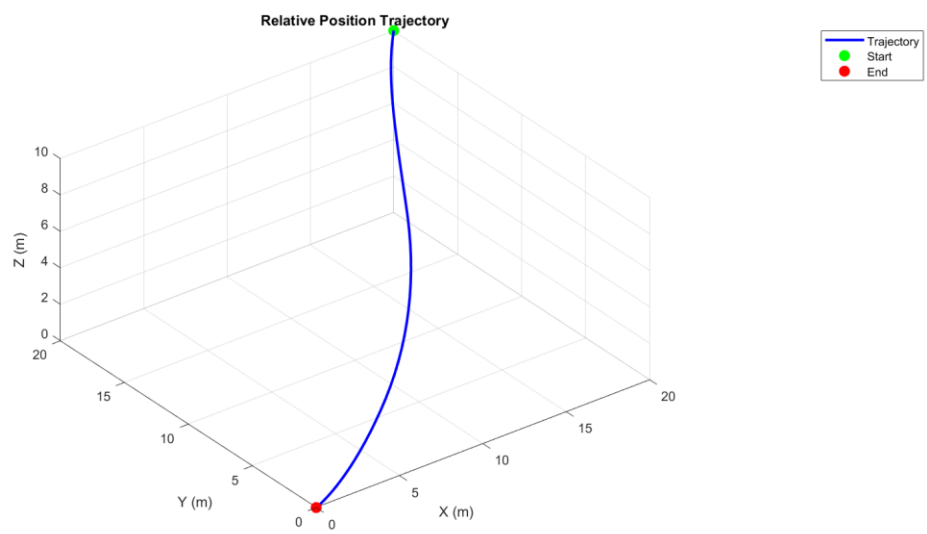
The results section will be set up very similarly to the previous one. There will be a heading for each different constraint where the results for them separately will be shown. As stated above, they will all share the same objective function and boundary conditions. The final section will be all of the constraints together, showing the feasibility of the entire problem of different values.

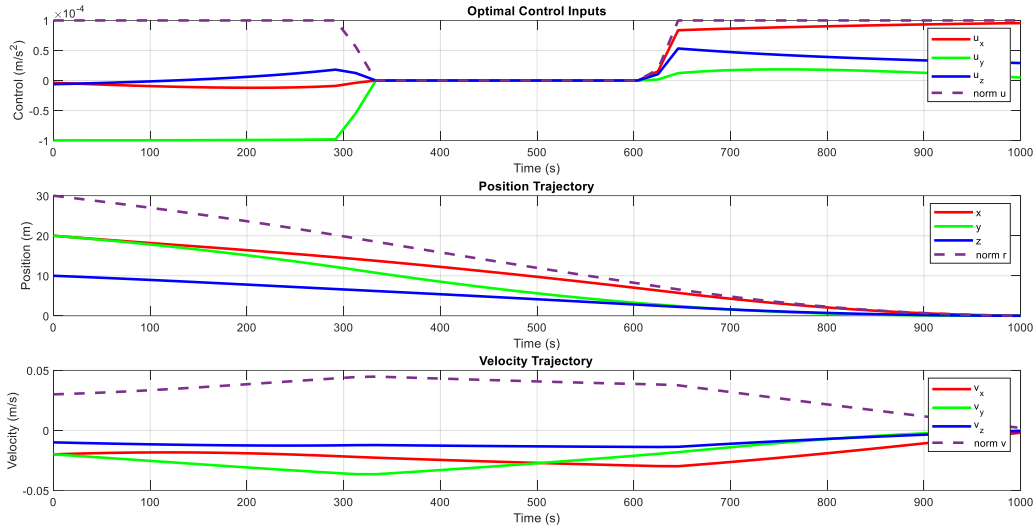
Shown below are the plots with no constraints, just boundary conditions, objective function, and dynamics.



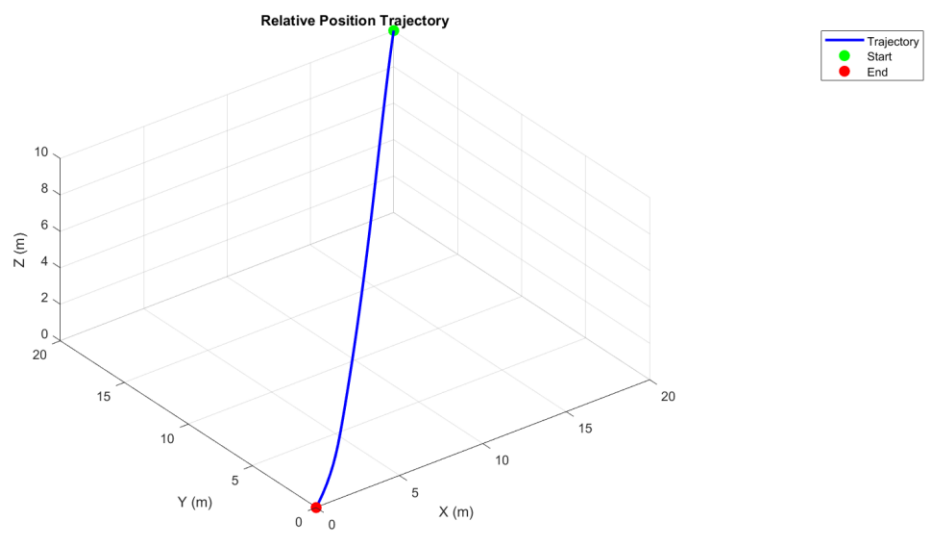


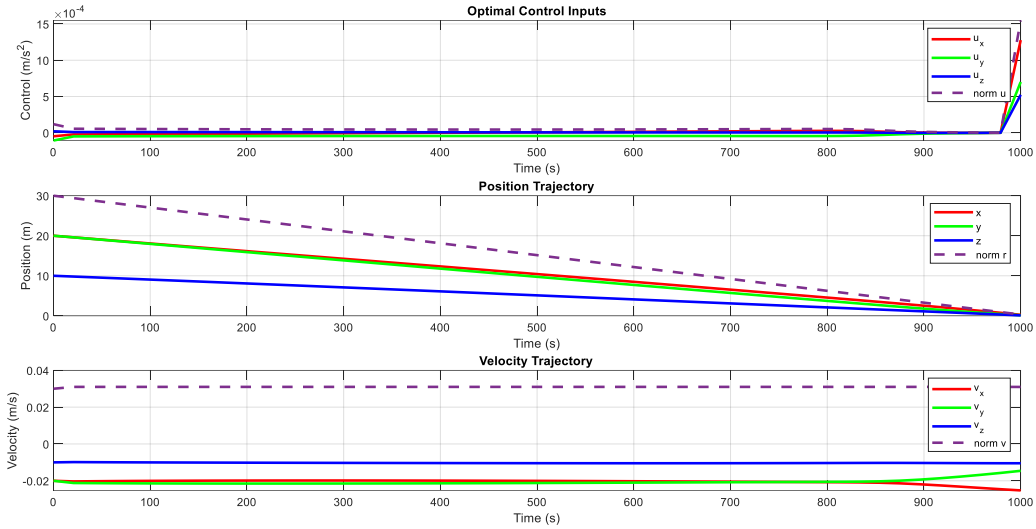
4.1 Maximum Thrust Constraint



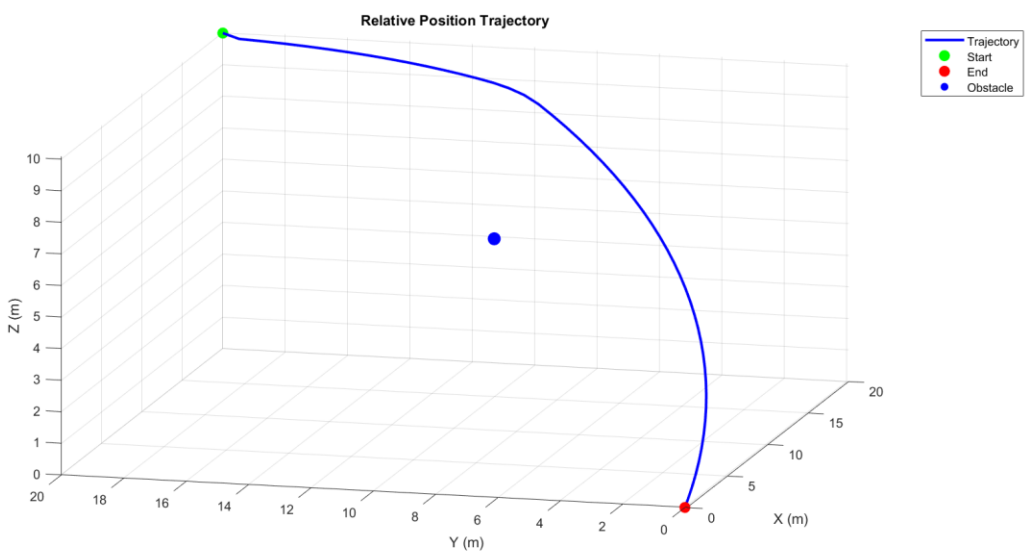


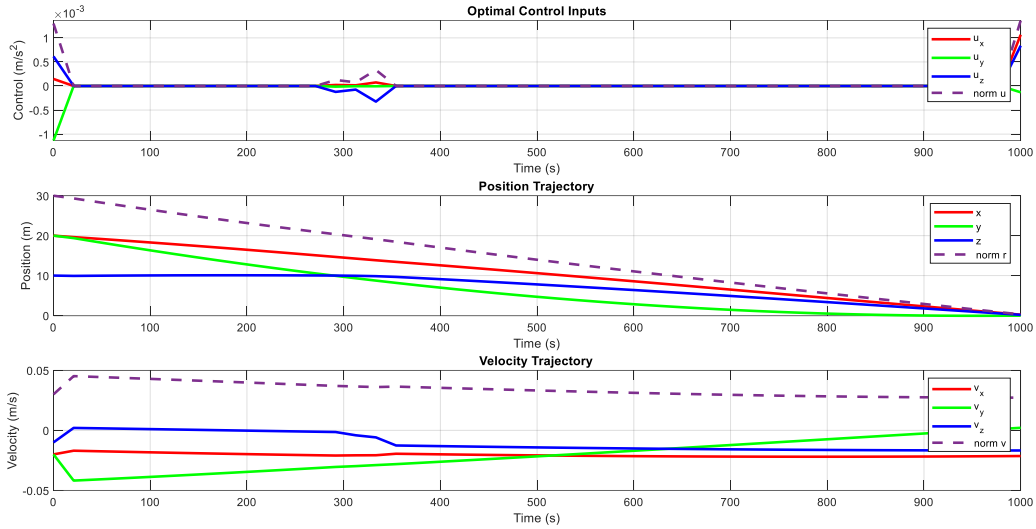
4.2 Maximum Velocity Constraint





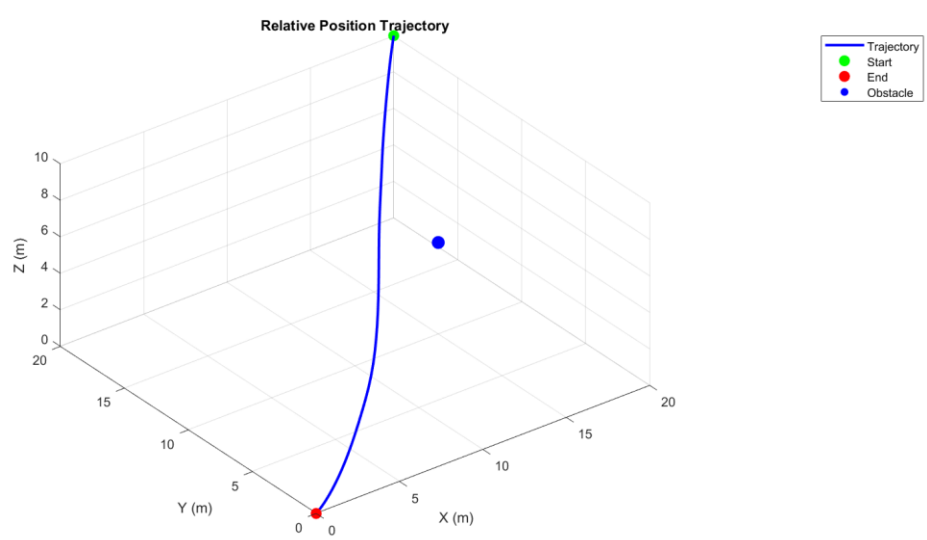
4.3 Collision Constraint

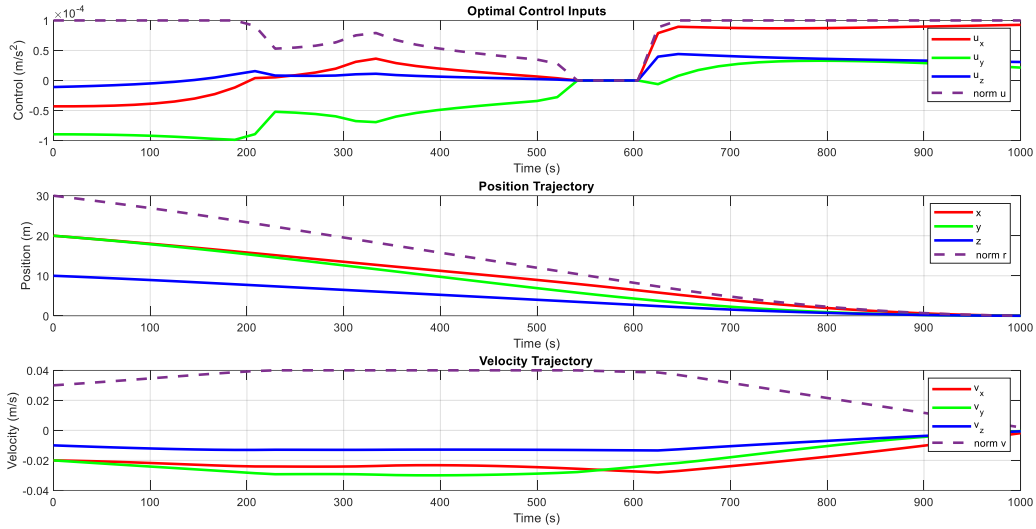




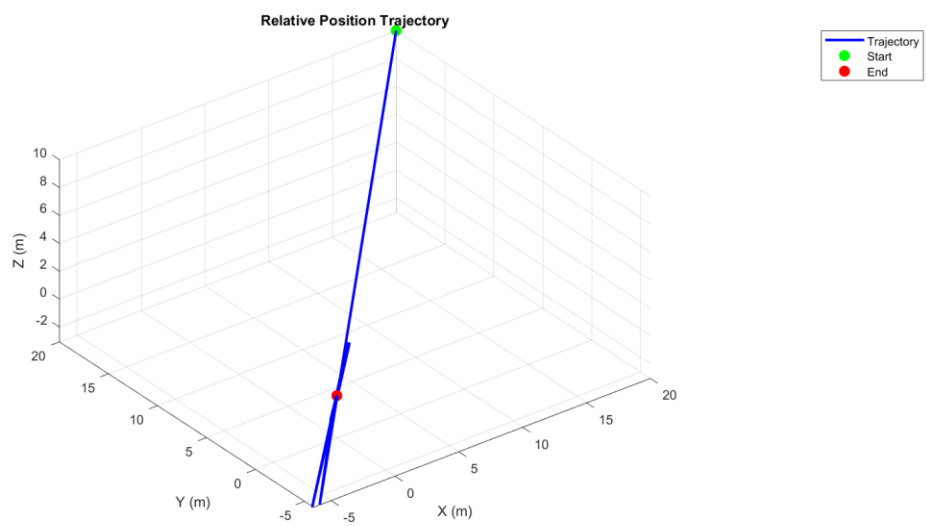
4.4 Top Three Constraints Together

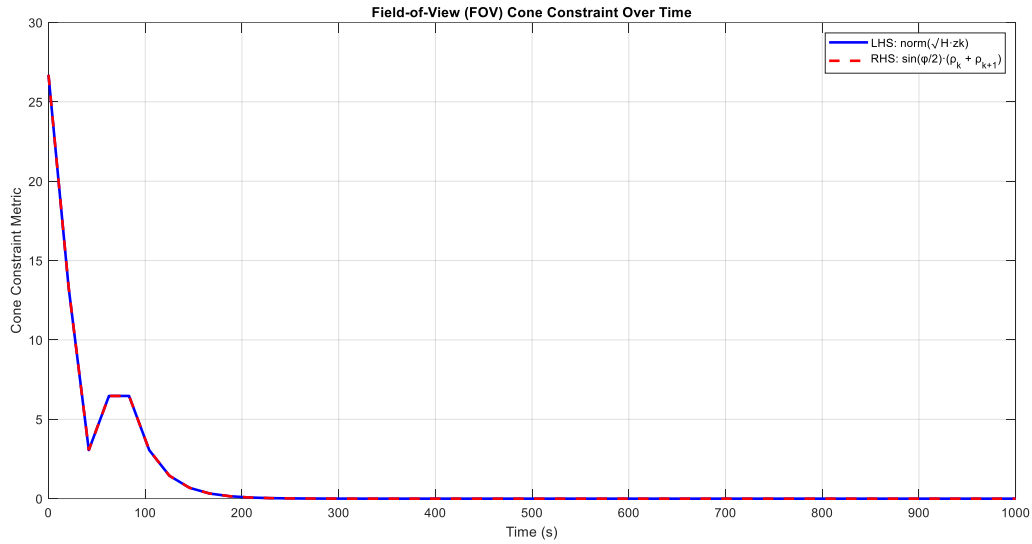
This problem will now look at the results for these three constraints combined. There have been some reworked variables for the constraints in order to make this problem feasible. This is because when there is an object in the path, there might need to be more thrust applied when time is kept constant. The new variables umax = 1e-4, vmax = 0.04, rsafe = 3, and r_obs = [15; 10; 5].





4.5 Field of View Constraint





4.6 All Constraints Combined Attempt

No graph results, infeasible.

5. Analysis of Results

5.1 Maximum Thrust Constraint

This is just a maximum thrust constraint that has been seen in class. The difference is the distance at which it operates as it is much closer. This leads to a different path than has been seen in class, when comparing the path with the max constraint to without, it is more direct but still has a slight curve. This is most likely because it has to slow down while using minimum thrust. It is notable for that curve and how it uses less thrust as the thrust cap is below the normal path with no constraints.

5.2 Maximum Velocity Constraint

This constraint is just velocity, no thrust. It is interesting because it takes a more direct path than the max thrust and no constraint path. This constraint was made purposely to cut off the velocity at a point that it does get up to in the no constraint path so it will actually change its course. This resulted in the path shown in the results and is most likely due to the fact that it takes the most direct path to still try to use minimum fuel while limiting velocity.

5.3 Collision Constraint

The collision constraint only result is extremely interesting. This one is similar to the others because there had to be an object placed specifically in the path, which matches how the constraint had to be placed to affect the original. This one was done using a convex analysis with no constraints and then a collision constraint convex analysis. This explains the path that the spacecraft takes and why it swings out. Something extremely interesting to note is that it avoids the object similarly to the nominal path but is a little more straight edged at the turn. The other notable thing is the control input graph that is still pretty low but is much more constant the whole time than the others. This makes sense at is constantly trying to avoid that object, it is still very interesting.

5.4 Top Three Constraints Together

This result is with all three of the previous constraints combined. This one was infeasible with the previous numbers for the constraints from the other results. This resulted in new numbers that had to be plugged in. This shows that having the three constraints together results in a more difficult problem that is infeasible in more cases. However, the resulting graph with the new constraints is interesting. The control plot is maxed out almost the whole time with only a little time spent at zero or almost zero. The velocity plot is cut off at its max point in the middle of the plot showing that it has to use extra thrust to not hit its max. As well as it moves around the object in order not to hit it. All in all, this is an extremely interesting result and can be useful in docking where there is another object in the path as well as a max velocity or thrust that has to be maintained for safety. This means that these constraints can be very useful in the future and this was a good level two goals to achieve.

5.5 Field of View Constraint

The field of view constraint is interesting because it is feasible in the sense of following the constraint but not physically feasible. Part of this is due to the fact that there is no collision constraint on the International Space Station so it just goes right through. It could also not be feasible due to the field

of view not being as robust due to not having extra constraints. This part will be discussed more below. The other interesting thing of note is the graph where it shows that feasibility of the constraint and that it is always within thirty degrees field of view. It starts out slightly facing away then faces toward the ISS and stays that way. The other thing is that it does that even though it goes in and out of the ISS, so there might be something a little off with the implementation but all in all it is pretty good. Also, there is a notable direct path that the spacecraft takes for this constraint as well. With more time I would refine this constraint alone, but the main goal would be to get it to all work together like the end level two goal as will be discussed in section 5.6.

5.6 All Constraints Combined Attempt

I attempted to do the field of view constraint with the other three, but this was extremely difficult and it did not work. This would require completely redoing the problem like they did in Sow. There needs to be a complete change of variables in order to make it more robust and feasible. I tried to implement it but it would end up with many more constraints that feasible for the timeframe and the fact that it is a one-man team. It would be extremely insightful with more time and manpower to continue work with this project.

6. Conclusion

This project has been extremely insightful into some of the important work that is done in these fields. Reading those research papers and attempting to replicate some of those results and adapt it for this paper was extremely difficult but rewarding. A key takeaway is the importance of carefully balancing constraint tightness and trajectory smoothness. Overly aggressive constraints can induce oscillations or infeasibility. Another take away is that some constraints cannot be made alone as they need others to be robust enough to handle the problems given. This is to say that there might be solutions out there but they are much harder to find without doing the entire problem, as can be seen in sections 4.5 and 5.5. I think an extremely important thing to pay attention to when moving forward is

that field of view constraint and when moving forward make sure it is visualized and robust. I really learned how to understand the convexification process and how CVX works much better.

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